CSC 251 HOMEWORK THREE

HEIGHT CONVERSION TABLE

THE ASSIGNMENT

For this assignment you will write code to produce a table that converts heights in feet and inches to centimeters. You should prompt your user to enter the range of values that they wish for the table to span: first a starting value for the height in feet and inches, then the ending value in feet and inches, assuming the table spans from a shorter height to a larger height. Give your user information concerning the range of values to be entered and allow them to enter either an integer or float value so that your code does not break easily. Round their values to the nearest integer using Python3's rounding rules.

Your program should then output a well-aligned and formatted table to span from the beginning height up to and including the ending height, incrementing 1 inch per line. The table header is provided in a file called `header.py`. Import this file as a module in your program and call the `header()` function. The height in feet and inches in the first two columns should be output with no floating-point values while the centimeter conversion should be output to two decimal values. All numeric values should be place-value aligned.

SAMPLE INPUT AND OUTPUT

Sample Run 1: Suppose your user enters starting height values of 4 feet, 10 inches and ending values of 6 feet, 1 inch. Your table output should match the values below, with an overall table width of 37 characters.

```
        Height Conversion Table
   feet    |   inches   ||     cm
=====================================
          4 |         10 ||     147.32
          4 |         11 ||     149.86
          5 |          0 ||     152.40
          5 |          1 ||     154.94
          5 |          2 ||     157.48
          5 |          3 ||     160.02
          5 |          4 ||     162.56
          5 |          5 ||     165.10
          5 |          6 ||     167.64
          5 |          7 ||     170.18
          5 |          8 ||     172.72
          5 |          9 ||     175.26
          5 |         10 ||     177.80
          5 |         11 ||     180.34
          6 |          0 ||     182.88
          6 |          1 ||     185.42
```

Sample Run 2: Suppose your user enters starting height values of 5 feet, 2 inches and ending values of 5 feet, 2 inches. Your table output should output only the single row of values.

```
        Height Conversion Table
   feet    |   inches   ||     cm
=====================================
         5 |          2 ||     157.48
```

Sample Run 3: Suppose your user enters starting height values of 5 feet, 12 inches. Since the inch value is over 11, your program should prompt them to re-enter BOTH values; they enter 6 and 0. When prompted for the ending values, they enter 5 feet, 1 inch. Since the ending height is less than the starting height, your program should prompt them to re-enter BOTH ending values; they enter 6 feet, 8 inches. In summary, the order of their input would be:

```
5 12 6 0 5 1 6 8
```

The following table should be output:

```
        Height Conversion Table
   feet    |   inches   ||     cm
=====================================
         6 |          0 ||     182.88
         6 |          1 ||     185.42
         6 |          2 ||     187.96
         6 |          3 ||     190.50
         6 |          4 ||     193.04
         6 |          5 ||     195.58
         6 |          6 ||     198.12
         6 |          7 ||     200.66
         6 |          8 ||     203.20
```

PROGRAM REQUIREMENTS

Your program should include the following features:

- A Python file named `hw3functions.py` that include at least the following functions:
  - A function called `get_values` that prompts the user for a height in feet and inches and returns the two acceptable values.
  - A function called `convert_values` that is passed two values, feet and inches, and returns the height in centimeters, assuming 2.54 cm per inch.
  - A function called `print_table_row` that prints out a single formatted row of values for the table.
- A `main` function file called `hw3main.py` that should import both the `header` and `hw3functions` modules and call their functions as necessary to gather user input and create the table.

- Other functions if you find them helpful, but they are not required.

- Clear user prompts. Points will be deducted for misspelled, unclear, and confusing prompts. For example, instead of:

  ```
  feet = input("enter feet")
  ```

  a better prompt would be (pardon the wrapping):

  ```
  feet = input("Enter the height in feet and inches. First enter the
  number of feet as an integer between 0 and 20: ")
  ```

  A strict format for the user prompts will not be provided for this assignment. In other words, they are not expected to match mine exactly.

- Input validation to test that the user's values are within range and that they don't enter ending values that are smaller than the starting values. Allow the user to enter both the feet and inches values before re-prompting them for BOTH values if one of the values is not reasonable. You may assume that reasonable values range from 0 feet, 0 inches up through values too large to be accommodated by the table format shown above. The values of feet and inches should be entered such that the number of inches does not exceed 11. In other words, "5 feet 15 inches" should be "6 feet 3 inches". If they enter 12 inches or more, prompt them to **reenter BOTH the number of feet and the number of inches**. If the ending values they enter are smaller than the starting values, prompt them to enter **both ending values again**. **Your code should not include exception handling** since this has not yet covered in class, so you may assume that the user enters numerical values, although they could be integers or float values. All float values entered should be rounded to integers.

- Table output that follows the formatting provided by the `header.py` file and the `table_output.txt` file provided.

- Documentation throughout both files that you write. Include an opening comment for each file, a comment for each function, and comments throughout your code. You do not need to document the `header.py` file that is provided for you.

- Global constants that replace magic numbers.

PROGRAM SUBMISSION

Zip together your three files: `header.py`, `hw3functions.py` and `hw3main.py` and submit the zipped file to Moodle. The zipped file should use the naming format: `hw3LastnameFirstInitial.` These naming conventions should be followed EXACTLY for full credit.